

Eman Hrustemović

Junior Data Scientist · Machine Learning · NLP

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[Portfolio](#) | github.com/EmanHrustemovic | linkedin.com/in/eman-hrustemovic

PROFESSIONAL SUMMARY

Recent IT graduate and first-year Master's student specialising in Data Science at International Burch University (IBU), Sarajevo. Experienced in building and validating machine learning models on large real-world datasets, with a focus on NLP, classification, and honest evaluation practices (leakage auditing, grouped validation). Proficient in Python, SQL, DuckDB, Pandas, and Scikit-learn. Currently completing the Machine Learning track at FlyRank AI, where a capstone research paper is deployed publicly and in mentor review. Eager to apply data science skills in a professional environment and contribute to data-driven projects.

EXPERIENCE

Machine Learning Intern — FlyRank AI, Remote

Jun 2026 – present

- Selected for the July 2026 ML cohort out of 20,000+ applicants from 100+ countries
- Built and validated a page-level content-decline scoring model on FlyRank's production search warehouse (79M+ rows); model beat a hand-written baseline rule by a wide margin (Precision@50: 0.70 → 0.96), re-validated under a stricter client-grouped split to rule out data leakage
- Assigned each of 92,148 scored pages a reason code and ranked action, turning model output into an editorial playbook with human-review rules and monitoring triggers
- Shipped a full capstone research paper — deployed publicly at emanhrustemovic.github.io/ml-capstone-paper; capstone submitted and currently in mentor review

PROJECTS

Content Decline Scoring — FlyRank ML Internship Capstone

2026

Python · DuckDB · Scikit-learn · Hugging Face Datasets

- Problem: content teams cannot manually review every page every month; needed a way to flag declining pages worth prioritizing for refresh
- Contribution: designed the full pipeline solo — data contract and leakage checks, a hand-written CTR-vs-position baseline, a Logistic Regression model, and an honest client-grouped re-validation
- Result: model reached Precision@50 of 0.96 vs. a 0.70 baseline, holding steady under a stricter honest split (AUC 0.649 vs. baseline 0.544) — a measured, reproducible lift, not a cherry-picked number

Multi-Task Sentiment Analysis on Amazon Food Reviews

2025

Python · Scikit-learn · HuggingFace · Conference paper, under review

- Problem: benchmark classical ML against transformer models for multi-task sentiment prediction on real, imbalanced review data
- Contribution: analysed 568,454 Amazon reviews; benchmarked Logistic Regression, Random Forest, k-NN, Naive Bayes, and DistilBERT across three classification tasks; applied SMOTE, TF-IDF (5,000 features), and 3-fold cross-validation
- Result: Logistic Regression reached 70.85% accuracy on rating prediction, ANN ~93% on reviewer ID, Random Forest ~70% ROC-AUC on helpfulness prediction

"Moje Zdravlje" — Full-Stack Health Web Application

2024 – 2025

PHP · REST API · SQL · HTML/CSS/JS · JWT · Graduation project

- Problem: build a working health-tracking platform end-to-end as a graduation capstone

- Contribution: built the PHP backend with a RESTful API and SQL database, JWT-based authentication, and a full CRUD interface; managed the complete SDLC from requirements to deployment
- Result: fully functional, deployed full-stack application demonstrating backend, frontend, and database integration

TECHNICAL SKILLS

Core DS/ML: Python, SQL, Pandas, NumPy, Scikit-learn, Jupyter Notebook

Data & Warehousing: DuckDB, Hugging Face Datasets, SQL aggregation over large-scale (79M+ row) production data

Model Validation: Grouped/stratified splits, leakage auditing, Precision@K evaluation

NLP & AI: TF-IDF, SMOTE, DistilBERT (HuggingFace), text preprocessing

Dev & Tools: Git, GitHub, Docker, VS Code, PyCharm, IntelliJ IDEA

Web & Other: PHP, HTML/CSS/JavaScript, REST API, JWT, Java, Kotlin

Productivity: Microsoft Office (Excel, Word, PowerPoint)

EDUCATION

MSc Information Technology — Data Science

2025 – present

International Burch University (IBU), Sarajevo

BSc Information Technology

2021 – 2025

International Burch University (IBU), Sarajevo

- Relevant coursework: Machine Learning, Databases, Software Engineering, OOP, Software Testing
- Co-authored the IBU Python Workbook (2023–2024), structured beginner-to-advanced exercises covering OOP, algorithms, and data structures, mentored by Prof. Dr. Dzelila Mehanovic

CERTIFICATIONS

AI Fluency for Builders — Anthropic & CodePath.org, 2026 — [Verify](#)

AI Fluency: AI Capabilities & Limitations — Anthropic, 2026 — [Verify](#)

AI Fluency: Framework & Foundations — Anthropic, 2026 — [Verify](#)

Claude 101 — Anthropic, 2026 — [Verify](#)

Introduction to Claude Cowork — Anthropic, 2026 — [Verify](#)

Machine Learning Track — FlyRank AI, 2026 — *certificate expected September 2026*

LANGUAGES

Bosnian: Native

English: Proficient — C1/C2

German: Elementary — A1/A2